

FTGP Group 11: Week 24 Sprint Report

EcoPassEU — Privacy-Aware Digital Product Passport DApp

Sprint Duration: 28 April 2026 – 4 May 2026

Report Date: 30 April 2026

University of Bristol

Last Sprint Review and Retrospective

Sprint Duration: 21 April 2026 – 27 April 2026

This section summarises the sprint review meeting held on 28 April 2026, marking the close of Sprint 2. The meeting was attended by all five team members in person: Luxiao Cao, Akshansh Rajora, Tianwei Yu, Detai Dong, and Fuyu Cao.

Completed Work

The following tasks were completed during Sprint 2, together with the primary contributors responsible for each item.

All sprint artefacts committed to GitHub

All members

All work produced during Sprint 2 was committed to the group's GitHub repository under the organisation account shared with supervising teaching staff. This includes revised paper drafts, updated smart contract code and ABI files, frontend deployment assets, Hardhat test outputs, and expanded data pipeline scripts. The repository is accessible at https://github.com/DeTaiDong/FTGP2425_Group1_11.

Paper writing: introduction, background, and design documentation *Luxiao Cao (lead)*

Luxiao Cao continued his leading role in academic writing this sprint, completing and refining the introduction and background sections carried over from Sprint 1, and producing a full design documentation chapter covering the system architecture of EcoPassEU. The design documentation details the off-chain storage model using IPFS and the Pinata pinning service, the on-chain ERC-1155 token structure and its relationship to Content Identifier (CID) storage, the JSON-LD schema used for Digital Product Passport records, and the interaction model between the React frontend and the Polygon Amoy testnet via Ethers.js. Luxiao Cao additionally conducted a systematic literature survey, identifying and integrating references across blockchain-based provenance systems, the EU Ecodesign for Sustainable Products Regulation (ESPR) framework, and Life Cycle Assessment (LCA) methodologies. All citations were formatted in compliance with the Springer LNCS style agreed by the group.

Hardhat testing

Luxiao Cao

In addition to his paper writing responsibilities, Luxiao Cao authored and executed a Hardhat test suite covering the deployed ERC-1155 smart contract. The suite tests all principal contract entry points—`mintBatch`, `updatePassportHash`, and the CID retrieval

utilities—verifying correct event emission, access-control enforcement, reversion on invalid inputs, and gas consumption under representative call patterns. All tests passed against the revised contract deployment. The results were committed to the repository and will serve as a primary data source for the gas cost analysis reported in the evaluation section of the paper.

Frontend: deployment and supplementary features

Detai Dong (lead)

Detai Dong completed the production deployment of the React frontend and delivered both supplementary features deferred from Sprint 1. The application was deployed via Vercel, with environment variables for the Polygon Amoy RPC endpoint and the deployed contract address injected at build time; a `vercel.json` configuration was committed to the repository to document the deployment parameters. The supplementary features delivered this sprint are: an export-to-PDF function for individual passport records, implemented using a client-side rendering approach to preserve layout fidelity without server-side dependencies; and a comparative product view enabling users to display two Digital Product Passport records side by side for direct comparison. Both features were integrated into the existing routing structure and verified against the live deployment without regressions to the core passport retrieval or QR code flows.

Smart contract: audit and optimisation

Akshansh Rajora

Akshansh Rajora conducted a comprehensive review of the deployed ERC-1155 contract, identifying and implementing several gas optimisation opportunities. Changes applied include elimination of redundant storage reads within view functions, consolidation of emitted event parameters to reduce per-transaction log data volume, and tightening of access-control modifiers to remove unnecessary conditional branching. NatSpec documentation was added to all public and external functions in accordance with Ethereum development conventions. The unit test suite was updated to achieve full branch coverage. The revised contract was redeployed to the Polygon Amoy testnet; the updated contract address and ABI were committed to `contracts/deployments/`.

Data pipeline: extended dataset and ESPRESSO integration

Fuyu Cao (lead),

Tianwei Yu

Fuyu Cao and Tianwei Yu expanded the working dataset from approximately 500 to 1,000 records, applying the filtering and normalisation pipeline established in Sprint 1 to the additional records and resolving a small number of previously unencountered UTF-8 encoding variants in the source CSV. Fuyu Cao led the ESPRESSO V2 integration effort, researching the characterisation factor tables, mapping emission factor identifiers to the existing product categories in the Open Beauty Facts dataset, and implementing the field population logic in the data pipeline script. Each passport record in the output now carries `co2Equivalent` and `waterStressIndex` fields populated from the ESPRESSO tables. Tianwei Yu took responsibility for the batch minting pipeline, coordinating the upload of expanded JSON-LD records to IPFS via Pinata and the subsequent minting of ERC-1155 tokens on Polygon Amoy, verifying that all 1,000 CIDs were correctly stored on-chain and resolvable. Total pinning bandwidth usage remained within the free-tier limits of the Pinata service. Fuyu Cao updated the data gap log to document all imputation decisions made during the expansion, including the justification for fallback values applied to records with missing packaging or ingredient fields.

Sprint coordination and product backlog management

Tianwei Yu

In his capacity as Product Owner for Sprint 2, Tianwei Yu maintained and prioritised the product backlog throughout the sprint, facilitating daily asynchronous standups and ensuring that dependencies between the data pipeline, smart contract, and frontend workstreams were surfaced and resolved promptly. He coordinated the mid-sprint realignment required when the ESPRESSO field mapping introduced a minor schema update, confirming with Akshansh Rajora that the on-chain ABI remained compatible and that no contract redeployment was required at that stage. Tianwei Yu additionally prepared the integration test plan used at the close of the sprint, specifying the verification criteria for the full 1,000-record end-to-end flow, and ran the integration test alongside Fuyu Cao to confirm successful resolution of all minted tokens via QR code in the deployed frontend.

Preliminary evaluation data collection

Fuyu Cao, Tianwei Yu

Fuyu Cao and Tianwei Yu initiated the evaluation suite in the final days of the sprint, collecting a preliminary set of measurements ahead of the formal evaluation planned for Sprint 3. Fuyu Cao ran hash integrity verification across a representative sample of 200 records from the expanded dataset, confirming that the SHA-256 digest computed client-side from IPFS-retrieved JSON-LD matched the value stored on-chain for every record in the sample. Tianwei Yu recorded ten QR code resolution latency samples under typical campus network conditions, capturing the time from QR code scan to full passport data render in the frontend. These preliminary results were committed to the repository as a structured CSV and will be extended to the full 1,000-record dataset in Sprint 3.

Changes and Incomplete Tasks

Methodology, results, and evaluation sections — deferred to Sprint 3.

The paper sections covering methodology, quantitative results, and formal evaluation were not completed within Sprint 2. Whilst gas cost data and preliminary QR latency and hash integrity measurements were collected, the full evaluation suite was not finalised in time to support complete paper drafting. These sections constitute the critical-path deliverable for Sprint 3.

Comprehensive evaluation suite — partially completed.

Gas cost measurements (via Hardhat), a preliminary hash integrity sample (200 records, Fuyu Cao), and ten QR resolution latency samples (Tianwei Yu) were collected during the sprint. Full hash integrity verification across all 1,000 records and frontend load time profiling remain outstanding and will be completed as the first priority of Sprint 3.

Retrospective

Start Doing	Stop Doing	Keep Doing
Draft results and evaluation sections incrementally as measurement data becomes available, rather than waiting for the complete dataset before beginning to write.	Treating paper writing as a strictly sequential activity that must follow all engineering work; writing and evaluation can proceed concurrently once component interfaces are stable.	Daily asynchronous standups, which continued to surface blockers promptly and keep all members aligned across parallel workstreams throughout the sprint.
Allocate a time-boxed evaluation window early in the sprint so that measurements are available to inform paper drafts throughout the sprint rather than only at its close.	Deferring evaluation planning until individual components reach full stability; smoke-level measurements can commence as soon as a component is in a testable state.	Feature branch discipline and pull-request review, which continued to provide a lightweight quality gate for both codebase and paper draft changes, and reduced merge conflicts noticeably.
Prepare a formal submission checklist at the outset of Sprint 3, covering code freeze, paper formatting compliance, repository archival, and submission portal requirements.		Clear task ownership and proactive cross-team communication, which maintained sprint velocity despite the breadth of simultaneous engineering, writing, and evaluation workstreams.

Next Sprint Planning

Sprint Duration: 28 April 2026 – 4 May 2026

The sprint planning meeting for Sprint 3 was held on 28 April 2026. The full team was present in person.

Product Owner

Tianwei Yu

Scrum Master

Detai Dong

Sprint Vision

Sprint 3 is the final sprint of the project. By the close of this sprint, the team aims to deliver a complete, submission-ready project paper and a fully evaluated, stable prototype. The principal deliverables are: the methodology, results, and evaluation sections of the paper; the full evaluation suite executed across the 1,000-record dataset; a finalised and proofread document compiled in Springer LNCS format; and a confirmed production deployment with the repository in an archival state accessible to the examiners. No new features will be added to the prototype this sprint; all engineering effort is directed at evaluation, documentation, and submission preparation.

Sprint Backlog

The following tasks have been selected from the product backlog for Sprint 3. Estimates are given in story points (1 point $\approx$ half a day of focused work).

Task	Description	Assignee(s)	Points
Paper writing: methodology, results & evaluation	Draft and finalise the methodology, results, and evaluation sections of the project paper, incorporating gas cost data, QR latency measurements, frontend load time figures, and hash integrity results. Produce all required tables and figures in Springer LNCS format.	Luxiao Cao (lead)	5
Comprehensive evaluation suite	Complete the full evaluation suite: hash integrity verification across all 1,000 records, QR code resolution latency (10 samples), gas cost per DPP issuance (mean and variance over 20 transactions), and frontend load time under representative network conditions. Extend preliminary data collected in Sprint 2 to the full dataset.	Fuyu Cao, Tianwei Yu (lead), all members	4
Conclusion and abstract	Write the paper conclusion synthesising the principal findings from the evaluation; revise the abstract to reflect the final scope and results of the project.	Luxiao Cao (lead)	2
Paper finalisation and proofreading	Integrate all sections into a single compiled document in Springer LNCS format; resolve cross-reference, citation, and formatting inconsistencies; conduct a full proofreading pass to ensure consistent British English spelling and academic register throughout.	Luxiao Cao, Fuyu Cao	3
Final deployment and repository archival	Verify the production deployment resolves all 1,000 records correctly; confirm all contract addresses, ABIs, and environment configuration files are current; archive the repository in the state required by the examiners.	Detai Dong, Akshansh Rajora	2
Total story points this sprint:			16

Notes on Member Availability

All five team members are available full-time in Bristol for the duration of Sprint 3. Given that this is the final sprint and the submission deadline is imminent, the team has

agreed that all engineering effort will be focused exclusively on evaluation and deployment verification; no further feature development will be undertaken. Luxiao Cao has reserved the final two days of the sprint for paper compilation, formatting, and proofreading. Fuyu Cao will assist with proofreading and reference verification in addition to leading the remaining hash integrity evaluation runs. Tianwei Yu will coordinate the full evaluation suite as Product Owner and will lead the QR latency and load time measurement sessions. Detai Dong and Akshansh Rajora will be available throughout the sprint to support any evaluation measurements requiring live application access.

Anything Else to Share

The GitHub repository continues to serve as the single source of truth for all project artefacts. All code, data pipeline scripts, test outputs, deployment configurations, and paper drafts from Sprints 1 and 2 are committed and accessible to supervising teaching staff at https://github.com/DeTaiDong/FTGP2425_Group1_11.

IPFS pinning usage following the expansion to 1,000 records remains within the Pinata free-tier bandwidth limits. Deployment of a self-hosted IPFS node is no longer considered necessary for the remainder of the project.

The end-to-end integration test conducted at the close of Sprint 2 by Fuyu Cao and Tianwei Yu confirmed that all 1,000 records are correctly pinned to IPFS, minted as ERC-1155 tokens on Polygon Amoy, and resolvable via QR code in the deployed React frontend. The prototype is stable and meets the requirements for the final evaluation and paper submission.